THE LECA RESURRECTION PROTOCOL

Recovery of the Last Eukaryotic Common Ancestor from Living Developmental Programs — One-Page Summary

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Every eukaryotic organism carries the Last Eukaryotic Common Ancestor (~1.5 Ga) as the base state of its developmental program. Placing any eukaryote in conditions geometrically derived from that attractor — before the experiment runs — stabilises it there. This is deliberate de-extinction of the oldest organism on Earth and the formal definition of a universal structural invariant observable in every reproductive cycle of every eukaryote alive — confirmed by a $200–400 experiment at any BSL-1 community biology laboratory in days. Pre-registered 2026-03-12.

The Theorem

The developmental trajectory from zygote to adult IS the evolutionary trajectory of the lineage, compressed into developmental time (the Reproduction Theorem). Therefore:

E **Evolutionary grade** — the attractor the lineage occupied at that point in its history.

D **Developmental position** — the equivalent stage in the developmental program today.

$E \equiv D$ (compressed). Arresting at position D stabilises grade E. The LECA sits at D=0: before the first multicellular step, in every eukaryote alive. **Conditions derived before the experiment. Prediction locked before the experiment. Result confirms the geometry.**

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The Three Pre-Registered DOIs

Document	DOI
Pre-Registration v1.0	...18986790
Registered Amendment A1	...18989573
The Ribosome Is the LUCA	...18988844

Repository: github.com/Eric-Robert-Lawson/ attractor-oncology

The Five Pre-Registered Predictions

P1 Arrest of any eukaryote before multicellular organisation produces a single-cell, nucleated, viable LECA-grade organism. Arm A alone.

P2 Arrested organisms from animal, plant, and fungal starting material are *morphologically indistinguishable* under light microscopy.

P3 *Genomically distinct* by sequencing — same attractor, different substrates.

P4 LECA-grade conserved genes active; all lineage-specific post-LECA genes silenced.

P5 Blind panel cannot identify starting kingdom from morphology alone (above 1/3 chance).

What Arm A Alone Establishes

Arm A is not a dormancy experiment. The arrest medium is derived geometrically from the LECA attractor before the experiment: Precambrian ionic composition, no nitrogen, microaerophilic (<3% O_2, pre-GOE), 20–22 °C.

Arm A confirmed = two simultaneous results:

1. De-extinction. First deliberate, pre-specified recovery of the LECA as an intentional target. $225 M has not produced a living mammoth (extinct 4,000 yr). Arm A recovers a 1.5 Ga organism for $200–400 in days — the oldest target is the easiest to recover.

2. Universal structural invariant defined. Developmental arrest is the instrument that formally defines a structure already present and observable in every reproductive cycle of every eukaryote on Earth — without intervention. Every developmental dataset ever collected is retroactively a measurement of this invariant. Every future developmental study confirms it. The LUCA runs continuously in every mitochondrion inside every LECA-grade cell right now. Once the base state is formally defined, the invariant is visible everywhere.

The Three Experimental Arms

Arm	Start	Cost	Timeline
A — Fungal	*S. cerevisiae*	$200–350	24–72 h
B — Animal	Mouse zygote	$400–600	24–72 h
C — Plant	*Arabidopsis*	$200–400	48–72 h

Arm A: P1 confirmed. Invariant base state defined. BSL-1. No credentials required. Arms A+B+C: morphologically identical across kingdoms, genomically distinct. Invariant confirmed across maximum phylogenetic distance.

Arm A — What You Actually Do

Day 0–3: Sporulate *S. cerevisiae* in 1% KAc, 25 °C, 48–72 h. Confirm asci by microscopy.

Day 3: Wash into arrest medium: 10 mM NaCl, 5 mM KCl, 2 mM $MgCl_2$, 1 mM $CaCl_2$, 0.1 mM $FeSO_4$, 1 mM KH_2PO_4, 0.1% glucose, **no nitrogen**, pH 6.5–7.0, 20–22 °C, <3% O_2.

Days 4–10: Observe 24 h, 48 h, 72 h. Phase contrast (40×) + DAPI. Pass: ≥80% single-cell, nucleus confirmed, ≤10% germination, viable = **P1 confirmed.**

Key control: At 72 h add nitrogen to half the culture — germination resumes, proving stabilisation not cell death.

Equipment: Microscope, incubator, laminar flow. All at ChiTownBio, Chicago.

The geometry came first. The invariant was always running. The experiment is the first deliberate observation. The LECA was never lost. It was never not there.